Intelligent Data Provenance Tracking: Blockchain-Enabled Trust Framework for MLOps Pipelines

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Abstract

As machine-learning (ML) models increasingly drive critical decisions, ensuring the trustworthiness, traceability, and integrity of data within Machine-Learning Operations (MLOps) has become a major challenge. Data provenance—the complete lineage of data from origin to model consumption—is often fragmented across distributed storage, cloud pipelines, and versioning systems. This paper proposes an intelligent, blockchain-enabled provenance framework that secures data lineage, enhances transparency, and supports automated compliance verification in MLOps environments. The system integrates a permissioned blockchain ledger with AI-assisted metadata extraction, enabling tamper-proof recording of dataset origins, transformations, feature derivations, and model usage events. Smart contracts enforce validation policies for data access, versioning, and model deployment, while provenance graphs visualize dependencies across components. Using Hyperledger Fabric and MLflow as the experimental base, the framework demonstrates 98 % provenance completeness and near-zero immutability breaches under simulated adversarial conditions. Performance evaluation shows minimal latency overhead (< 6 %) when integrated into CI/CD pipelines. The proposed approach advances responsible and auditable MLOps, ensuring that data artifacts can be reliably traced for regulatory, ethical, and security purposes. By combining distributed ledger technology with intelligent metadata reasoning, the framework establishes a new paradigm of trust-by-design automation in AI systems.

Keywords

Data Provenance · Blockchain · MLOps · Smart Contracts · Data Lineage · Trust Automation · Model Governance · Responsible AI

1. Introduction

The evolution of MLOps has transformed how organizations develop, deploy, and monitor machine-learning models. However, as ML pipelines automate data ingestion, feature engineering, training, and deployment, ensuring data traceability and accountability has become increasingly complex. Inconsistencies in dataset versions, undetected data tampering, and opaque preprocessing pipelines can result in biased or unreliable models—issues that are particularly critical in domains such as healthcare, finance, and autonomous systems. Data provenance tracking aims to capture the full lineage of data: where it originates, how it evolves, and where it is consumed. In traditional systems, this is achieved through metadata logging and version control (e.g., MLflow, DVC). Yet, such mechanisms rely on centralized storage that is vulnerable to manipulation and lacks verifiable integrity. Blockchain technology offers a

promising solution by providing an immutable, decentralized ledger capable of recording every transaction or modification in a verifiable manner. When integrated with AI-driven metadata extraction and smart-contract-based governance, blockchain enables automated and transparent provenance tracking across distributed MLOps components.

This paper introduces a Blockchain-Enabled Data Provenance Framework (BDPF) that leverages blockchain for secure lineage recording and AI models for intelligent metadata enrichment. The system is designed to (1) prevent tampering, (2) ensure compliance with data governance standards (e.g., GDPR, ISO/IEC 25012), and (3) enhance the trustworthiness of ML models through verifiable audit trails.

The remainder of the paper reviews related work, details the architecture and methodology of BDPF, presents experimental results, and discusses its implications for sustainable and responsible AI operations.

2. Literature Review

Early provenance systems relied on relational metadata repositories and workflow engines such as **Taverna**, **Kepler**, and **ProvONE** (Buneman et al., 2001). These systems captured workflow dependencies but lacked immutability and decentralized verification.

With the rise of big-data and ML ecosystems, several solutions—**Apache Atlas**, **MLflow**, and **TensorFlow Extended (TFX)**—introduced lineage tracking integrated with metadata stores. While these tools improved traceability, they remained **centralized** and **trust-dependent**: administrators could still modify or delete historical records.

Recent studies have explored **blockchain-based provenance** for scientific data and IoT systems. Ramachandran et al. (2019) proposed a blockchain ledger for scientific workflows, ensuring verifiable experiment records. Similarly, Liang et al. (2020) designed a smart-contract-driven data-sharing protocol enabling access control and integrity validation. In the AI domain, Li and Huang (2021) highlighted blockchain's potential for model-artifact governance but noted challenges in scalability and real-time throughput.

AI-enhanced provenance extends this paradigm by applying machine-learning models to enrich provenance graphs—automatically inferring relationships among datasets, transformations, and models (Zhao et al., 2022). Combining this with blockchain creates a **dual-layer trust model**: AI provides semantic understanding, while blockchain guarantees tamper resistance.

Despite promising developments, few implementations integrate these technologies directly within **MLOps pipelines**. Current gaps include (1) lack of standardized metadata schemas across ML workflows, (2) high computational cost of blockchain consensus, and (3) absence of intelligent validation mechanisms.

This paper addresses these gaps through a **permissioned blockchain with AI-driven metadata extraction**, optimizing performance while ensuring end-to-end lineage integrity. The proposed approach contributes to the growing discourse on **trustworthy, auditable, and explainable MLOps**.

3. Methodology

The proposed **Blockchain-Enabled Data Provenance Framework (BDPF)** consists of four interlinked layers (Figure 1):

1. **Data Ingestion & Metadata Extraction Layer:**

- Incoming datasets are automatically profiled for attributes such as source, schema, timestamp, and ownership.
- A transformer-based NLP model (e.g., BERT) extracts semantic tags and contextual lineage information from dataset documentation and code comments.

2. **Provenance Recording Layer (Blockchain):**

- Uses **Hyperledger Fabric** as a permissioned ledger to record cryptographic hashes of data assets and model artifacts.
- Each block represents a provenance event (e.g., data transformation, feature extraction, model deployment) signed by authorized nodes.
- **Smart contracts (Chaincode)** enforce version control and access validation rules.

3. **Intelligent Verification Layer:**

- Employs a graph-neural-network (GNN) model to infer missing lineage links and detect anomalies or unregistered data movements.

4. **Visualization & Governance Layer:**

- Provenance graphs are displayed via Neo4j dashboards; compliance rules (e.g., GDPR "right to be forgotten") are validated through smart contracts.

Workflow Integration:
The framework interfaces with MLflow and Kubeflow pipelines through APIs, ensuring that every dataset or model artifact processed in MLOps automatically generates a verifiable blockchain transaction.

4. Results

Experiments were conducted using synthetic and real MLOps workloads on a Hyperledger Fabric (v2.5) network with five peers and two orderers. Provenance data from MLflow tracking servers and Kaggle data-transformation logs were streamed into the blockchain.

- **Provenance completeness:** 98 % of lineage events were successfully captured and verified.
- **Tamper detection:** All 1,000 injected adversarial modifications were identified (100 % detection).
- **Latency overhead:** Average transaction delay = 142 ms (≈ 5.7 % slowdown relative to baseline pipelines).
- **Storage efficiency:** Block compression reduced ledger growth by 22 %.

These results confirm that integrating blockchain into MLOps pipelines can secure provenance tracking without significantly affecting operational performance.

5. Discussion

The evaluation demonstrates that blockchain-enabled provenance tracking provides verifiable and tamper-resistant data lineage across distributed ML workflows. The permissioned architecture of Hyperledger ensures scalability and controlled access, addressing privacy and throughput limitations common in public blockchains.

The integration of AI-based metadata extraction adds a semantic layer to provenance tracking—automatically linking datasets to models and transformations without manual annotation. This hybrid approach reduces human error, enhances discoverability, and supports compliance audits.

Nevertheless, several challenges persist. First, ledger scalability remains constrained by block-generation latency, particularly under heavy CI/CD loads. Techniques such as off-chain storage and hash-anchoring could alleviate this. Second, ensuring interoperability with diverse MLOps tools requires adherence to metadata standards like W3C PROV-DM or OpenLineage. Third, explainability of AI-inferred provenance edges must be improved to gain auditor trust.

Incorporating federated provenance sharing across multi-tenant organizations is a promising direction. By combining zero-knowledge proofs with blockchain, sensitive data can remain private while still contributing to verifiable lineage. Overall, the BDPF framework demonstrates a practical path toward trust-centric, compliance-ready AI pipelines.

6. Conclusion

This paper presented an intelligent blockchain-enabled data-provenance framework for MLOps pipelines. The integration of AI-based metadata reasoning with an immutable distributed ledger ensures end-to-end traceability, integrity, and accountability of datasets and model artifacts. Experimental results confirm high provenance completeness and minimal performance overhead, validating the framework's feasibility for enterprise-grade deployments.

By embedding trust-by-design principles into MLOps, the framework strengthens auditability, supports regulatory compliance, and mitigates risks of data tampering or unverified model usage. Future research should explore lightweight consensus protocols, standardized provenance ontologies, and explainable graph reasoning to enhance scalability and transparency. The convergence of AI and blockchain thus marks a pivotal step toward trustworthy, autonomous, and ethically governed AI operations.

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